Protecting Quantum Information: BB84, Stabilizer Codes and a Memory-Assisted Key Budget

Verified revision (v2) — Aaditri Saraswat. Every number below is produced by the code in this repository (`python run_all.py`) with exact density-matrix simulation; the test suite (`python -m pytest tests`) passes.

1. What was checked, and the verdict

Claim in v1 report	Status	Verified value / note
BB84 QBER = p_eve/4 under intercept-resend	Confirmed	gate-level QBER 0.250 at p_eve=1; 0.100 at 0.4 (theory 0.100). Expected value is exactly 0.25; v1's plot point (about 0.27) and its text ("never reaches 25%") are both finite-sample noise.
BB84 QBER = 2p/3 under channel noise	Confirmed	0.198 at p=0.30 (theory 0.200)
Shor–Preskill threshold 11.0%	Confirmed, wording fixed	0.1100. It is where this proof's *one-way* rate 1–2h(Q) hits zero; v1's "no protocol can extract key above it" is too strong (two-way post-processing tolerates more).
Decoy-state max range 166 km	Confirmed	165.9 km (my parameters: f=1.16, q=1/2, v1 did not state them)
"Naive analysis overstates range by 16 km"	**Reframed**	The naive formula is insecure, not a competing analysis (183.4 km). The right comparison is secure-without-decoy (GLLP, best μ per distance): **66 km**. Decoy states buy about 100 km.
37–39% of clicks are multi-photon	Confirmed	37.0–39.2% for 0–150 km at μ=0.5
3-qubit codes beat unencoded up to p=0.5 under depolarizing noise	**Not supported as stated**	True only for pure bit-flip noise on the protected inputs (closed form matched to 1e-9). Under depolarizing noise averaged over all four logical inputs they are *worse* than no code at every p tested (p=0.05: 0.0484 vs 0.0333), because each protects only one Pauli type.
Shor code beats unencoded below p≈0.10–0.12	Confirmed	break-even p = 0.120 (p = total Pauli error probability per qubit; unprotected error = 2p/3)
Bit-flip/phase-flip crossover at T2≈79 μs (T1=80 μs)	Confirmed	78.9 μs
(implicit) the better code is worth choosing	**Qualified**	Neither 3- nor 5-qubit repetition code beats an unprotected qubit at any T2 tested; the crossover is real but both lose.
5-qubit code worse than 3-qubit	Confirmed (direction)	T2=160 μs: bit-flip 3q 0.0208 vs 5q 0.0328; CX count 4 vs 8 here (v1 used ancilla syndromes: 8 vs 16)
Steane purification stops helping at p≈0.09–0.10	Confirmed (exact)	p* = 0.0969 by exact enumeration; v1's table entries at p=0.10 are sampling noise (4,000 shots).
…which is "close to the 11% BB84 threshold"	**Misleading**	p is the depolarizing parameter, not the QBER. In QBER terms the purifier stops helping at **6.5%**, well below 11%. The 'coincidence' disappears once units match.

2. Method (what makes the numbers trustworthy)

Exact, not sampled: QEC circuits run on Aer's density-matrix simulator and read exact outcome probabilities (no shot noise, no error bars needed). BB84 is simulated transmission by transmission as real circuits (Eve = mid-circuit measure, reset, conditional re-preparation). **Verified before use:** every code corrects every single protected error at every location for all four logical inputs (X/Z for repetition codes, X/Y/Z for Shor); closed forms $3q^2(1–q)+q^3$ and the 5-qubit analogue are reproduced to 1e-9. **Four-input averaging:** a repetition code protects one Pauli type, so |0>,|1>,|+>,|–> are averaged; for BB84 memory this average *is* the added QBER, because BB84 uses two bases. **Noise:** Pauli (X,Y,Z each p/3) or thermal relaxation on every gate and on the storage window (CX 300 ns, 1q 50 ns). Decoding uses perfect classical majority logic.

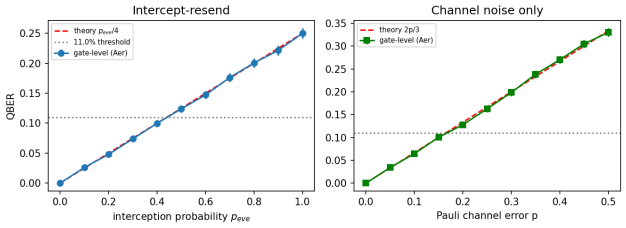


Fig. 1 Gate-level BB84 against theory (95% bars).

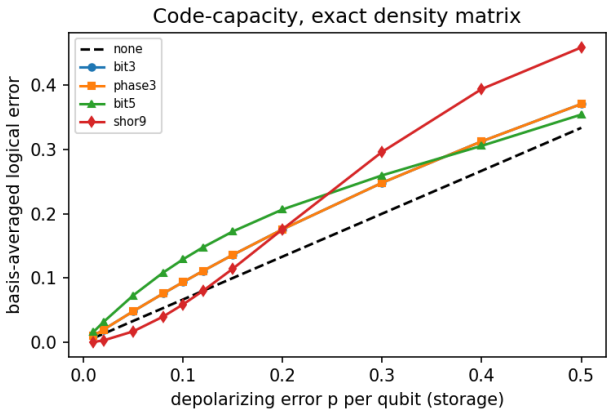


Fig. 2 Code-capacity: only Shor beats unencoded, and only at small p.

3. Novel part: a memory-assisted key budget

The v1 report proposes (next steps) combining BB84 with a QEC-protected node. Here that is done. A received qubit waits t before measurement (delayed sifting, repeater or memory node). Memory-induced QBER $e_{mem}(t)$ is combined with channel QBER as independent flips, $Q = Q_{ch}+e_{mem}-2Q_{ch}e_{mem}$, and compared with the 11.0% threshold.

Coherence	Shor-9 beats unprotected between	t_{max} unprotected (Q_{ch}=3%)	t_{max} Shor-9 (Q_{ch}=3%)	Key-rate gain inside window
T1=T2=500 µs	8–45 µs	92 µs	72 µs	at t=30 µs: 0.365 → 0.403 bits/sifted bit
T1=T2=1000 µs	7–104 µs	184 µs	145 µs	at t=30 µs: 0.480 → 0.536 bits/sifted bit
T1=T2=5000 µs	7–561 µs	919 µs	762 µs	at t=30 µs: 0.583 → 0.603 bits/sifted bit

Finding. Encoding does not extend how long a key can survive storage: the Shor code helps only inside a finite window (below it, encode/decode gates cost more than they save; above it, multi-qubit errors beat the code) and that window closes *before* the 11% line. What it does buy is a higher key rate at moderate storage times where error is already small. Two practical rules follow: the encoded memory only pays off when T_1,T_2 are several hundred µs or better, and the storage time must sit inside the window.

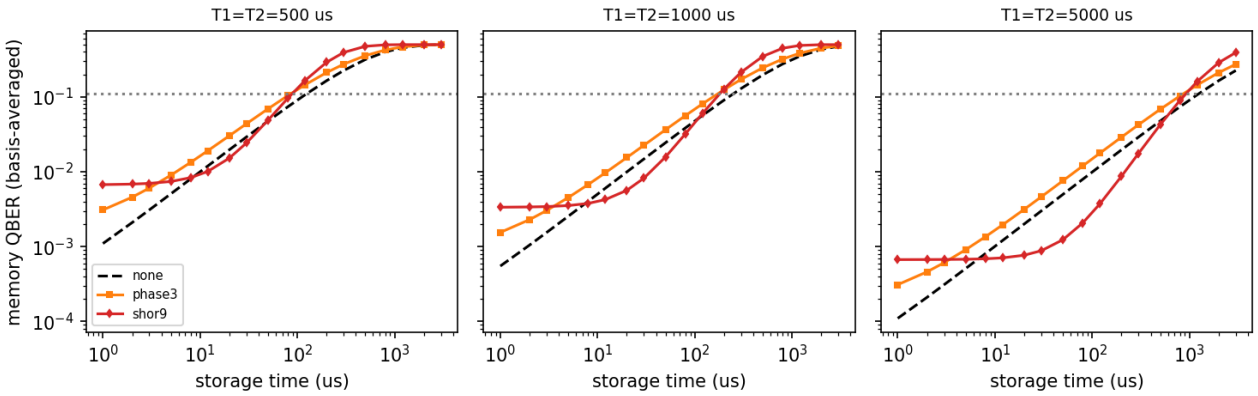


Fig. 3 Memory QBER vs storage time for unprotected, 3-qubit phase-flip and Shor-9 memories; dotted line = 11% threshold. Shor-9 wins only inside a window.

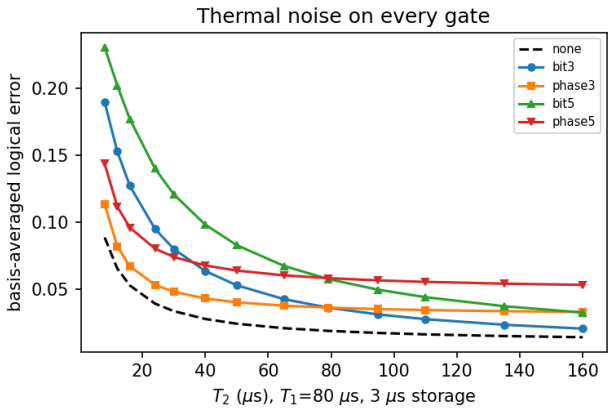


Fig. 4 Thermal noise on every gate (T1=80 µs): every code loses to unprotected storage; bit/phase codes cross near 79 µs.

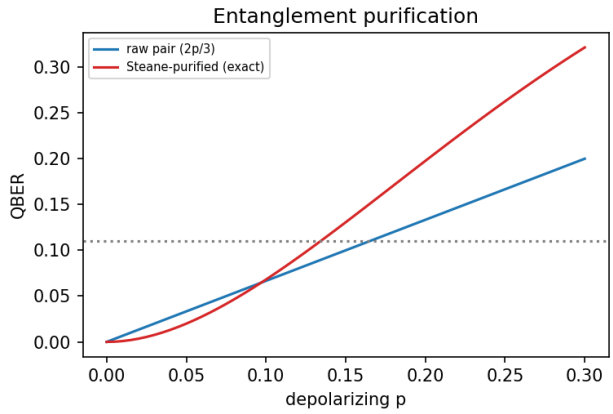


Fig. 5 Exact Steane-purified QBER vs raw pair; pseudo-threshold p=0.097, i.e. raw QBER 6.5%.

4. Decoy states, finite size and purification

Without decoy states the secure range collapses to 66 km even with µ re-optimised at each distance, because Eve may own every multi-photon pulse; the decoy method restores 166 km. A finite-size model (illustrative, after Tomamichel et al. 2012 / Lim et al. 2014; not a certified bound) shows the asymptotic rate is unreachable for short blocks: the smallest block giving positive key is N = 4,168 (QBER 1%), 5,754 (QBER 2%), 8,317 (QBER 3%), 19,054 (QBER 5%), 63,095 (QBER 7%). The Steane purifier is evaluated by exact enumeration over all 128 error patterns (CSS decoding is independent for X and Z).

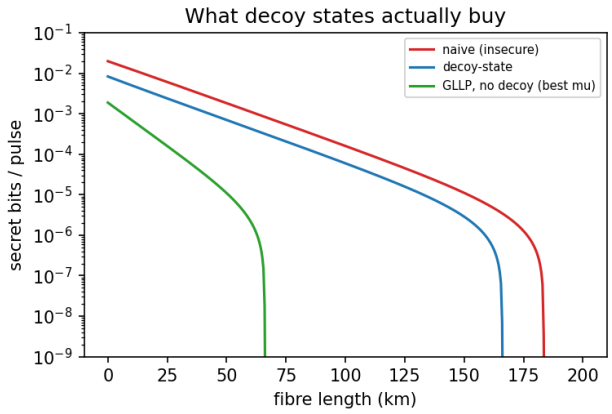


Fig. 6 Secure no-decoy range is 66 km; decoy states give 166 km.

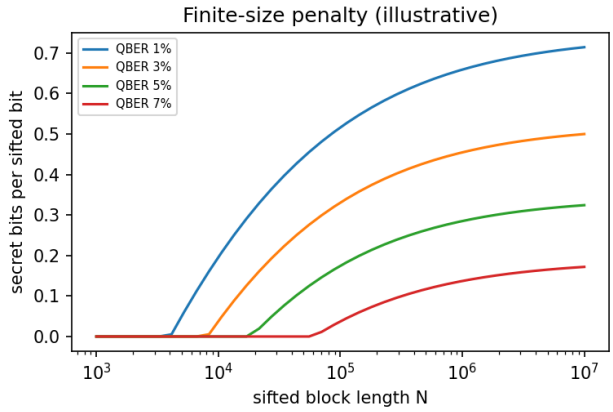


Fig. 7 Finite-size penalty on the key rate.

5. Limitations

• Aer is a classical simulator: its 'quantum' bits are pseudorandom, so nothing here is a security claim about real devices. A real-hardware run (IBM Quantum) is a separate step.

• Decoding is ideal classical majority logic with noise-free control; ancilla/measurement noise and repeated rounds are not modelled, which favours the codes. Even so, they lose in most regimes.

• Decoy analysis is asymptotic with infinitely many decoy pulses; the fibre/detector parameters and error-correction efficiency f are my choices (they reproduce v1's 166 km). The finite-key formula is a simplified illustration.

• State-preparation and final basis-change gates sit inside the noisy region for both protected and unprotected memories (a constant, tiny offset at the coherence times used). Thermal channel only (T1/T2); no crosstalk, leakage, or biased-noise-tailored codes (e.g. XZZX), which are the natural next step.